1.1. Divisibility.

<u>Definition</u> Let $a, b \in \mathbb{Z}$. Then a <u>divides</u> b ($a|b$) if $b = a \cdot c$ for some $c \in \mathbb{Z}$. (say a is a <u>divisor</u> or <u>factor</u> of b)

<u>Example</u> $3|6$ $(6 = 3 \cdot 2)$. $\pm 3 | \pm 6$.

$3 \nmid 5$ $(5 = 3 \cdot 5/3,\ 5/3 \notin \mathbb{Z})$

Let $a \in \mathbb{Z}$. Then $a|0$ $(0 = a \cdot 0)$ and $0|a \Leftrightarrow a = 0 \cdot c$ for some $c \in \mathbb{Z} \Leftrightarrow a = 0$.

<u>Proposition</u> Let $a, b, c \in \mathbb{Z}$. Then $a|b$ and $b|c$ implies $a|c$.

Proof. There exist $e, f \in \mathbb{Z}$ such that $b = a \cdot e$ and $c = b \cdot f$. So $c = bf = (ae)f = a(ef)$. Thus $a|c$. □

<u>Proposition</u> Let $a, b, c, m, n \in \mathbb{Z}$. If $c|a$ and $c|b$ then $c | (ma + nb)$.

Proof. There exist $e, f \in \mathbb{Z}$ such that $a = ce$ and $b = cf$. So $ma + nb = mce + ncf = c(me + nf)$. □

<u>Definition</u> Given $x \in \mathbb{R}$, define $\lfloor x \rfloor$ to be the greatest integer less than or equal to x.

<u>Example</u> $\lfloor 2 \rfloor = 2$, $\lfloor \frac{5}{3} \rfloor = 1$, $\lfloor -\frac{5}{3} \rfloor = -2$, $\lfloor \pi \rfloor = 3$.

* <u>Lemma</u> Let $x \in \mathbb{R}$. Then $x - 1 < \lfloor x \rfloor \leq x$.

Proof. By definition, $\lfloor x \rfloor \leq x$. To show $x - 1 < \lfloor x \rfloor$, assume $x - 1 \geq \lfloor x \rfloor$. Then $x \geq \lfloor x \rfloor + 1$ and thus $\lfloor x \rfloor$ is not the greatest integer less than or equal to x. This contradiction implies $x - 1 < \lfloor x \rfloor$. □

<u>Theorem</u> (Division Algorithm) Let $a, b \in \mathbb{Z}$ with $b > 0$. Then there exist unique $q, r \in \mathbb{Z}$ such that $a = bq + r$ with $0 \leq r < b$. (q: quotient, r: remainder)

Proof. Let $q = \lfloor \frac{a}{b} \rfloor$ and $r = a - bq$. Then $a = bq + r$. By the above Lemma, $\frac{a}{b} - 1 < q \leq \frac{a}{b}$. Thus $b - a > -bq \geq -a$. So $b > a - bq = r \geq 0$. This shows the existence of q and r.

To show the uniqueness, assume $a = bq_1 + r_1$, $0 \leq r_1 < b$ and $a = bq_2 + r_2$, $0 \leq r_2 < b$. Then $0 = a - a = b(q_1 - q_2) + r_1 - r_2$. Thus $r_2 - r_1 = b(q_1 - q_2)$ and $b | (r_2 - r_1)$. Since $0 \leq r_1 < b$ and $0 \leq r_2 < b$, we have $-b < r_2 - r_1 < b$. So $r_2 - r_1 = 0$, i.e., $r_2 = r_1$. Then $b(q_1 - q_2) = r_2 - r_1 = 0$. Thus $q_1 = q_2$. □

<u>Example</u> $a = -5$, $b = 3$. $-5 = 3 \cdot (-2) + 1$.

<u>Definition</u> An integer n is <u>even</u> if $2|n$ and n is odd if $2 \nmid n$.

<u>Example</u> $\{\cdots -6, -4, -2, 0, 2, 4, 6, \cdots\}$ even

$\{\cdots -5, -3, -1, 0, 1, 3, 5, \cdots\}$ odd.

Hwk: 1(a–d), 2, 3(a,b,d), 4, 10(a)

1.2 Prime Numbers

<u>Definition</u> Let $p \in \mathbb{Z}$, $p > 1$. Say p is <u>prime</u> if the only positive divisors of p are 1 and p. If $n \in \mathbb{Z}$, $n > 1$, and n is not prime, then n is said to be <u>composite</u>.

<u>Example</u>: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47 are primes.

<u>Lemma</u> Every integer greater than 1 has a prime factor.

Proof. Assume some integer $n > 1$ has no prime factor. Let n be as small as possible. Since $n|n$, n cannot be prime. So $n = ab$ for some integers $a, b > 1$. Then $b \geq 1$ implies $a < n$. So a has a prime factor p (since n is as small as possible). Then $p|a$ and $a|n$ implies $p|n$, a contradiction. □

<u>Theorem</u> (Euclid) There are infinitely many prime numbers.

Proof. Assume there are only finitely many prime numbers, say $p_1, p_2, \cdots, p_n$. The number $N = p_1 p_2 \cdots p_n + 1$ has a prime factor by the above Lemma. So $p_i | N$ for some i. Then $p_i | (N - p_1 p_2 \cdots p_n) = 1$, a contradiction. □

<u>Proposition</u> A composite number n must have a prime divisor $p \leq \sqrt{n}$.

Proof. There exist $a, b \in \mathbb{Z}$ such that $n = ab$, $1 < a < n$, $1 < b < n$. Assume $a \leq b$, without loss of generality. Then $a \leq \sqrt{n}$ (otherwise $b \geq a > \sqrt{n}$ implies $n = ab > \sqrt{n} \cdot \sqrt{n} = n$, a contradiction). Then a has a prime divisor p by the previous Lemma. We have $p \leq a \leq \sqrt{n}$ and $p | a | n \Rightarrow p | n$. □

<u>Example</u> (Sieve) Find primes ≤ 40: delete multiples of $2, 3, 5 \leq \sqrt{40}$.

~~2~~ ~~3~~ ~~4~~ ~~5~~ ~~6~~ ~~7~~ ~~8~~ ~~9~~ ~~10~~ ~~11~~ ~~12~~ 13 ~~14~~ ~~15~~ ~~16~~ 17 ~~18~~ 19 ~~20~~ 21 ~~22~~ 23 ~~24~~ ~~25~~ ~~26~~ ~~27~~ ~~28~~ 29 ~~30~~ 31 ~~32~~ ~~33~~ ~~34~~ ~~35~~ ~~36~~ 37 ~~38~~ ~~39~~ ~~40~~

<u>Proposition</u> For any $n \in \mathbb{Z}$, $n \geq 1$, there exist n consecutive composite numbers.

Proof. $(n+1)! + 2, (n+1)! + 3, \cdots, (n+1)! + n + 1$ are n consecutive composite numbers, since $i | (n+1)! + i$ for all $i = 2, 3, \cdots, n+1$. □

<u>Conjecture</u> (Twin Prime) There are infinitely many prime numbers p such that $p + 2$ is also prime. ($318032361 \cdot 2^{107001} \pm 1$, 2001)

<u>Theorem</u> (Prime Number Theorem) $\lim_{x \to \infty} \frac{\pi(x) \ln x}{x} = 1$, where $\pi(x) = |\{\text{primes} \leq x\}|$. (e.g., $\pi(40) = 12$, $\pi(50) = 15$).

<u>Conjecture</u> (Goldbach, 1742) Every even integer > 2 can be written as the sum of two prime numbers. ($4 = 2 + 2$, $6 = 3 + 3$, $8 = 3 + 5$, $\cdots$)

<u>Conjecture</u> There are infinitely many Mersenne primes ($2^p - 1$, p prime)

<u>Definition</u> A prime number of the form $2^p - 1$ with p prime is a <u>Mersenne prime</u>. A prime of the form $2^{2^n} + 1$ is a <u>Fermat prime</u>.

<u>Example</u> $3 = 2^2 - 1$, $7 = 2^3 - 1$, $31 = 2^5 - 1$ are prime, but $2^{11} - 1 = 23 \cdot 89$ is not.

$3 = 2^{2^0} + 1$, $5 = 2^{2^1} + 1$, $17 = 2^{2^2} + 1$, $257 = 2^{2^3} + 1$, $65537 = 2^{2^4} + 1$.

<u>Conjecture</u> There are exactly five Fermat primes.

<u>Conjecture</u> There are infinitely many prime numbers of the form $n^2 + 1$.

Hwk: 16(a,b), 18(a), 21(a,b), 22, 24*.

1.3. Greatest Common Divisors

<u>Definition.</u> Let $a, b \in \mathbb{Z}$ with a and b not both zero. The <u>greatest common divisor</u> of a and b, denoted (a,b), is the greatest positive integer d such that $d \mid a$ and $d \mid b$. If $(a,b)=1$ then a and b are <u>relatively prime</u> (or <u>coprime</u>). $(0,0)$ is undefined.

If $(a,b)=d$ then $(-a,b)=(a,-b)=(-a,-b)=d$.

<u>Example</u> $(24,60)=12$. $(24,0)=24$. $(24,35)=1$.

<u>Proposition.</u> Let $a,b \in \mathbb{Z}$ with $(a,b)=d$. Then $\left(\frac{a}{d}, \frac{b}{d}\right)=1$.

Proof. Let $\left(\frac{a}{d}, \frac{b}{d}\right)=d'$. Then $\frac{a}{d}=d' \cdot e$ and $\frac{b}{d}=d' \cdot f$. So $a=dd'e$ and $b=dd'f$. Thus dd' is a common divisor of a and b. But $d=(a,b)$ is the greatest common divisor of a and b. Hence $d'=1$. $\square$

<u>Proposition.</u> Let $a,b \in \mathbb{Z}$ with a and b not both zero. Then $(a,b)=\min\{ma+nb : m,n \in \mathbb{Z}, ma+nb>0\}$.

Proof. The set $S=\{ma+nb : m,n \in \mathbb{Z}, ma+nb>0\}$ is nonempty since $|a|+|b|=\pm a \pm b \in S$. Let $d=\min S$. By the division algorithm, $a=dq+r$, $0 \le r < d$. Then $r=a-dq=a-(ma+nb)q=(1-mq)a-nqb$. If $r \in S$ then $r<d$ is a contradiction. Thus $r \notin S$. This implies $r=0$. So $a=dq$ and $d \mid a$. Similarly, $d \mid b$. Let c be a common divisor of a and b. Then $c \mid (ma+nb)=d$. So $d=(a,b)$. $\square$

<u>Corollary</u> $(a,b)=1 \Leftrightarrow ma+nb=1$ for some $m,n \in \mathbb{Z}$.

Proof. Let $S=\{ma+nb : m,n \in \mathbb{Z}, ma+nb>0\}$.
$(\Rightarrow)$ $(a,b)=1 \Rightarrow 1 \in S \Rightarrow ma+nb=1$ for some $m,n \in \mathbb{Z}$.
$(\Leftarrow)$ $ma+nb=1 \Rightarrow 1 \in S \Rightarrow \min S=1$. $\square$

<u>Definition.</u> Let $a_1, a_2, \cdots, a_n \in \mathbb{Z}$, not all zero. The <u>greatest common divisor</u> of $a_1, a_2, \cdots, a_n$, denoted $(a_1, a_2, \cdots, a_n)$, is the greatest integer d such that $d \mid a_i$, $i=1,2,\cdots,n$. If $(a_1, \cdots, a_n)=1$ then $a_1, \cdots, a_n$ are <u>relatively prime</u>.

If $(a_i, a_j)=1$ for all pairs i,j with $i \neq j$ then $a_1, \cdots, a_n$ are <u>pairwise relatively prime</u>.

<u>Example</u>: $(24,60,30)=6$. $(24,60,49)=1$.
$24, 60, 49$ are relatively prime, but not pairwise relatively prime since $(24,60)=12 \neq 1$.

<u>Corollary</u> If $d \mid a$ and $d \mid b$ then $d \mid (a,b)$.

Proof. $(a,b)=ma+nb$ for some $m,n \in \mathbb{Z}$. Thus $d \mid a$ and $d \mid b \Rightarrow d \mid (ma+nb)=(a,b)$. $\square$

Hwk: 32 (b,c,d), 33(a,b), 34(a,b), 36, 37*

1.4 The Euclidean Algorithm

<u>Lemma</u> Let $a,b \in \mathbb{Z}$, $a \ge b > 0$, and $a=bq+r$ with $q,r \in \mathbb{Z}$. Then $(a,b)=(b,r)$.

Proof. Let c be a common divisor of a and b. Then $c \mid (a-qb)=r$. So c is also a common divisor of b and r. Conversely, if c is a common divisor of b and r then $c \mid (bq+r)=a$. So c is also a common divisor of a and r. Thus $(a,b)=(b,r)$. $\square$

<u>Theorem</u> (The Euclidean Algorithm). Let $a,b \in \mathbb{Z}$ with $a \ge b > 0$. There exist $q_1, r_1 \in \mathbb{Z}$ such that $a=bq_1+r_1$, $0 \le r_1 < b$. If $r_1>0$ then there exist $q_2, r_2 \in \mathbb{Z}$ such that $b=r_1q_2+r_2$, $0 \le r_2 < r_1$. If $r_2>0$ then there exist $q_3, r_3 \in \mathbb{Z}$ such that $r_1=r_2q_3+r_3$, $0 \le r_3 < r_2$. Continue this process. Then $r_n=0$ for some n. If $n=1$ then $(a,b)=b$. If $n>1$ then $(a,b)=r_{n-1}$.

Proof. Since $r_1 > r_2 > r_3 > \cdots \ge 0$, we have $r_n=0$ for some n. If $n=1$ then $b \mid a$ and thus $(a,b)=b$. By the above Lemma, if $n>1$ then $(a,b)=(b,r_1)=(r_1,r_2)=\cdots=(r_{n-1},r_n)=(r_{n-1},0)=r_{n-1}$. $\square$

<u>Example</u>

$803=154 \cdot 5+33$
$154=33 \cdot 4+22$
$33=22 \cdot 1+11$
$22=11 \cdot 2+0$

So $(803,154)=11$.

$(803,154)=11=33-22$
$=33-(154-33 \cdot 4)$
$=33 \cdot 5-154$
$=(803-154 \cdot 5) \cdot 5-154$
$=803 \cdot 5+154 \cdot (-26)$.

Hwk. 54(a,b), 56.

1.5. The Fundamental Theorem of Arithmetic

<u>Lemma</u> (Euclid) Let $a, b, p \in \mathbb{Z}$ with p prime.
If $p \mid ab$ then $p \mid a$ or $p \mid b$.

Proof. Assume $p \nmid a$. Then $(a,p)=1$. There exist $m, n \in \mathbb{Z}$ such that $ma+np=1$. Also, $p \mid ab$ implies $ab = pc$ for some $c \in \mathbb{Z}$. Then $b = mab + npb = mpc + npc$. Thus $p \mid b$. □

This lemma fails if p is composite. Thus [p is prime if and only if $p \mid ab \Rightarrow p \mid a$ or $p \mid b$.]
(This is the definition of prime in more general settings)

<u>Corollary</u>. Let $a_1, a_2, \dots, a_n \in \mathbb{Z}$ and p a prime. Then $p \mid a_1 a_2 \cdots a_n \Rightarrow p \mid a_i$ for some i.

Proof. We induct on n. The result is trivial if $n=1$. Assume it holds for $n=k$, i.e., $p \mid a_1 \cdots a_k \Rightarrow p \mid a_i$ for some $i \in \{1, \dots, k\}$. Then $p \mid a_1 \cdots a_{k+1}$ implies $p \mid a_1 \cdots a_k$ or $p \mid a_{k+1}$. If the latter holds then we are done. If the former holds then the induction hypothesis implies $p \mid a_i$ for some $i \le k$. □

<u>Theorem</u> (Fundamental Theorem of Arithmetic)
Every integer great than 1 can be written as $p_1^{a_1} \cdot p_2^{a_2} \cdots p_n^{a_n}$, with $p_1, p_2, \dots, p_n$ distinct primes and $a_1, a_2, \dots, a_n$ positive integers. This is the <u>prime factorization</u> of the integer, which is unique except for the arrangement of $p_1^{a_1}, \dots, p_n^{a_n}$.

Proof. Let k be the smallest integer that does not have a prime factorization. Then k is not prime. So $k = ab$, $1 < a < b$, $1 < b < k$. Then a and b have prime factorizations by the minimality of k. Thus $k = ab$ also has a prime factorization. This contradiction shows that every interger greater than 1 has a prime factorization.

To show the uniqueness, assume k has two prime factorizations $k = p_1^{a_1} \cdots p_n^{a_n} = q_1^{b_1} \cdots q_m^{b_m}$ with $p_1 < p_2 < \cdots < p_n$ and $q_1 < q_2 < \cdots < q_m$.
For each i, we have $p_i \mid k = q_1^{b_1} \cdots q_m^{b_m}$.
Thus $p_i \mid q_j$ for some j (Corollary 1.15).
Then $p_i = q_j$. Similary, for each j, we have $q_j = p_i$ for some i. Thus $n = m$ and $p_i = q_i$ for $i = 1, \dots, n$. Then $k = p_1^{a_1} \cdots p_n^{a_n} = p_1^{b_1} \cdots p_n^{b_n}$.
Assume $a_i \neq b_i$ for some i, say $a_i < b_i$. Then $p_i^{b_i} \mid k = p_1^{a_1} \cdots p_n^{a_n} \Rightarrow p_i^{b_i - a_i} \mid p_1^{a_1} \cdots p_{i-1}^{a_{i-1}} p_{i+1}^{a_{i+1}} \cdots p_n^{a_n}$.
$\Rightarrow p_i \mid p_j$ for some $j \neq i$, a contradiction. □

<u>Example</u>. $756 = 2 \cdot 378 = 2 \cdot 2 \cdot 189 = 2 \cdot 2 \cdot 3 \cdot 63 = 2 \cdot 2 \cdot 3 \cdot 7 \cdot 9 = 2^2 3^3 \cdot 7$

<u>Definition</u> Let $a, b \in \mathbb{Z}$, $a > 0$, $b > 0$. The <u>least common multiple</u> of a and b, denoted $[a,b]$, is the least positive integer m such that $a \mid m$ and $b \mid m$.

<u>Example</u> $[6,7] = 42$, $[6,8] = 24$. $[756, 2205] = 2^2 3^3 5^1 7^2 = 26460$

$756 = 2^2 3^3 5^0 7^1$, $2205 = 2^0 \cdot 3^2 5^1 7^2 \Rightarrow (756, 2205) = 2^0 3^2 5^0 7^1 = 63$.

<u>Proposition</u> Let $a, b \in \mathbb{Z}$, $a, b > 1$. If $a = p_1^{a_1} \cdots p_n^{a_n}$ and $b = p_1^{b_1} \cdots p_n^{b_n}$, with $p_1, \dots, p_n$ distinct primes and $a_1, \dots, a_n, b_1, \dots, b_n \ge 0$, then $(a,b) = p_1^{\min\{a_1, b_1\}} \cdots p_n^{\min\{a_n, b_n\}}$ and

$$[a,b] = p_1^{\max\{a_1, b_1\}} \cdots p_n^{\max\{a_n, b_n\}}.$$

Proof. This is obvious from the Fundamental Theorem of Arithmetics.

<u>Remark</u>. Prime factorizations are hard to compute in general. So Euclidean algorithm is still a good way to compute (a,b).

<u>Lemma</u> $\max\{x,y\} + \min\{x,y\} = x + y$.

<u>Theorem</u> Let $a, b \in \mathbb{Z}$ with $a, b > 0$. Then $(a,b)[a,b] = ab$.

Proof. Write $a = p_1^{a_1} \cdots p_n^{a_n}$ and $b = p_1^{b_1} \cdots p_n^{b_n}$. Then

$$(a,b)[a,b] = p_1^{\min\{a_1, b_1\}} \cdots p_n^{\min\{a_n, b_n\}} \cdot p_1^{\max\{a_1, b_1\}} \cdots p_n^{\max\{a_n, b_n\}}$$
$$= p_1^{\min\{a_1, b_1\} + \max\{a_1, b_1\}} \cdots p_n^{\min\{a_n, b_n\} + \max\{a_n, b_n\}}$$
$$= p_1^{a_1 + b_1} \cdots p_n^{a_n + b_n} = ab. \quad \square$$

<u>Example</u>. $[803, 154] = 803 \cdot 154 / (803, 154) = 803 \cdot 154 / 11 = 11242$.

<u>Corollary</u>. Let $a, b \in \mathbb{Z}$ with $a, b > 0$. Then $[a,b] = ab \Leftrightarrow (a,b) = 1$.

<u>Theorem</u> (Dirichlet's Theorem on Prime Numbers in Arithmetic Progression)
Let $a, b \in \mathbb{Z}$ with $a, b > 0$ and $(a,b) = 1$. Then $a, a+b, a+2b, \dots, a+nb, \dots$ contain infinitely many prime numbers. (Proof skipped).

<u>Lemma</u> Let $a, b \in \mathbb{Z}$. If $a = 4n+1$ and $b = 4m+1$ for some $n, m \in \mathbb{Z}$ then $ab = 4 \cdot k + 1$ for some $k \in \mathbb{Z}$.

Proof. $ab = (4n+1)(4m+1) = 16mn + 4m + 4n + 1 = 4(\overset{k}{4mn + m + n}) + 1$. □.

<u>Proposition</u> There are infinitely many primes of the form $4n+3$.

Proof. Assume there are only finitely many primes of the form $4n+3$, say $p_0 = 3, p_1, p_2, \dots, p_r$. Let $N = 4 p_1 p_2 \cdots p_r + 3$. If all prime factors of N are of the form $4n+1$ then N is of the same form, a contradiction. Thus there exists a prime $p = (4n+3) \mid N$ (Exercise 85).
If $p = p_0 = 3 \mid N$ then $3 \mid (N-3) = 4 p_1 \cdots p_r \Rightarrow 3 \mid p_i$ for some i, a contradiction.
If $p = p_i$, $i \ge 1$, then $p_i \mid (N - 4 p_1 \cdots p_r) = 3$, a contradiction. □

Hwk. 59 (abc), 60 (a,b,d), 61 (a), 63 (a), 85, 86.